Detection, Automation, and Behavioural Analysis of SSH Brute-Force Attacks in Linux Authentication Logs Using Python and Splunk Cloud SIEM

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Executive Summary

This project presents a comprehensive (SOC)-style investigation into suspicious authentication activity within Linux system logs, with a focus on identifying and analysing SSH brute-force attack patterns.

A publicly available dataset from GitHub (LogPai LogHub) was utilized to simulate real-world attack telemetry. The investigation began with manual log inspection in Visual Studio Code, where repeated occurrences of "authentication failure", "invalid user", and "user unknown" were identified. These patterns are well-established indicators of unauthorized access attempts, often associated with credential brute-force attacks and username enumeration techniques.

To enhance detection efficiency and reduce manual effort, a Python-based log parsing script was developed. This script automated the extraction and categorization of suspicious authentication events, enabling rapid identification of high-frequency failed login attempts and anomalous user activity. The output was structured into a dataset suitable for further analysis.

For scalable analysis and correlation, the raw log data was ingested into Splunk Cloud SIEM, where advanced querying and visualization were performed using SPL (Search Processing Language). The investigation revealed several critical findings:
The root account was the most frequently targeted user, indicating adversarial intent to gain privileged access.
Multiple remote hosts (rhost) were observed initiating authentication attempts, suggesting a distributed attack pattern.
The most active source IP generated 80 failed login attempts, a strong indicator of automated attack tooling.
Time-series analysis showed authentication attempts occurring within very short intervals (seconds), confirming non-human, automated behaviour consistent with brute-force scripts or botnet activity.
Visualization of event frequency highlighted distinct spikes in login attempts, further validating coordinated attack bursts.
From a threat intelligence perspective, the observed behaviour aligns with MITRE ATT&CK technique T1110 (Brute Force), specifically password guessing and password spraying variants.

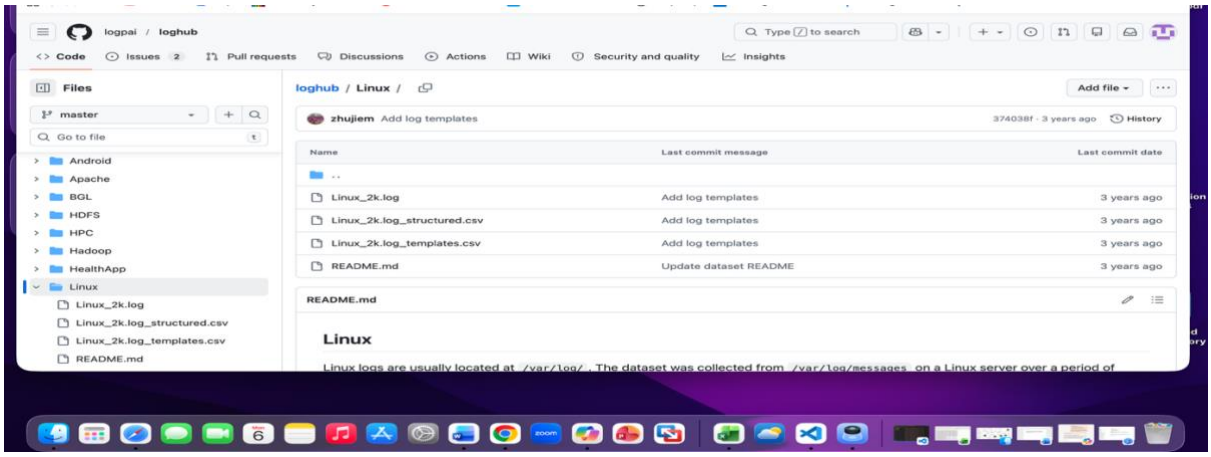
This project demonstrates practical, hands-on SOC analyst capabilities, including:

- Log-based threat detection
- Automation using Python for security analysis
- SIEM ingestion and SPL-based investigation
- Identification of attacker tactics, techniques, and patterns
- Data-driven security reporting and visualization

Overall, the findings confirm the presence of a coordinated brute-force attack campaign, likely automated, targeting privileged Linux accounts. The project highlights the importance of integrating log analysis, automation, and SIEM capabilities to effectively detect and respond to authentication-based threats.

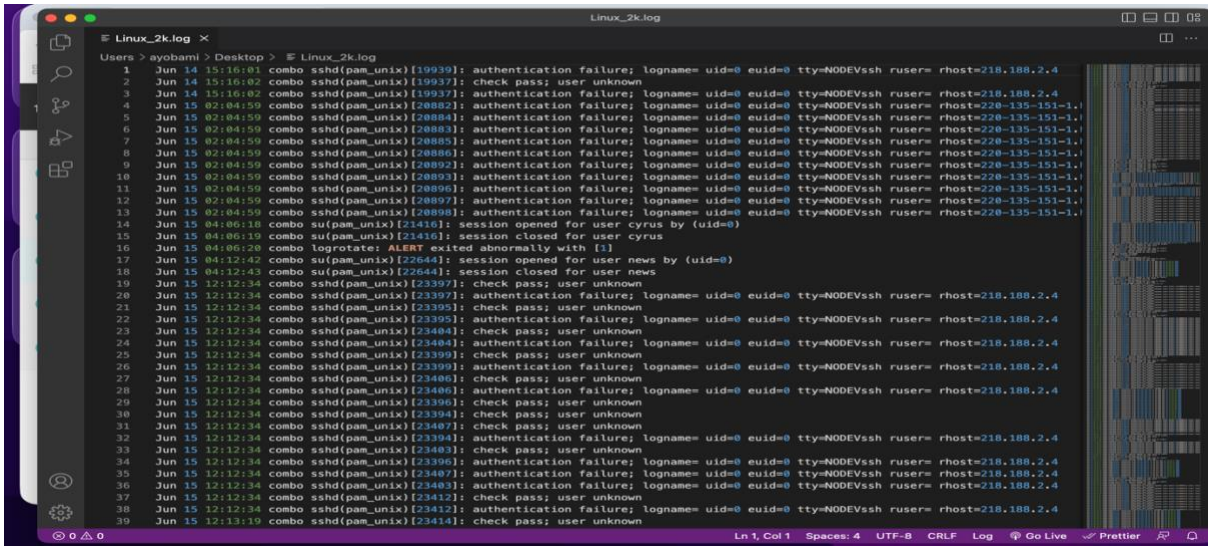
Investigating suspicious SSH authentication activity in Linux system logs to detect brute-force attacks.

The Public dataset was gotten from GitHub (LogPai LogHub) used to simulate real-world attack activity.



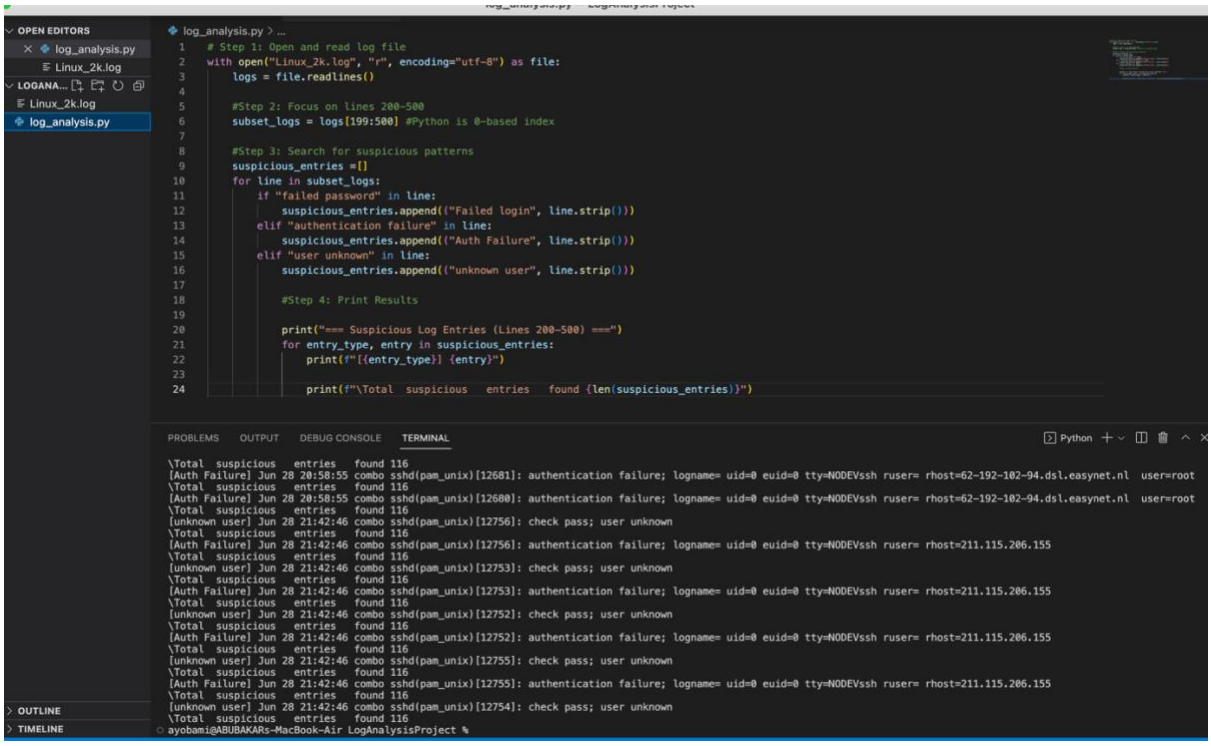
Manual Log Review:

- The Linux Logs was inspected manually using Visual Studio Code.
- The Key pattern found in the logs are “authentication failure,” “invalid user,” and “user unknown” indicating potential brute-force attempts.
- Multiple entries from the different (rhost)source IP suggested coordinated attacks targeting privileged account (root).

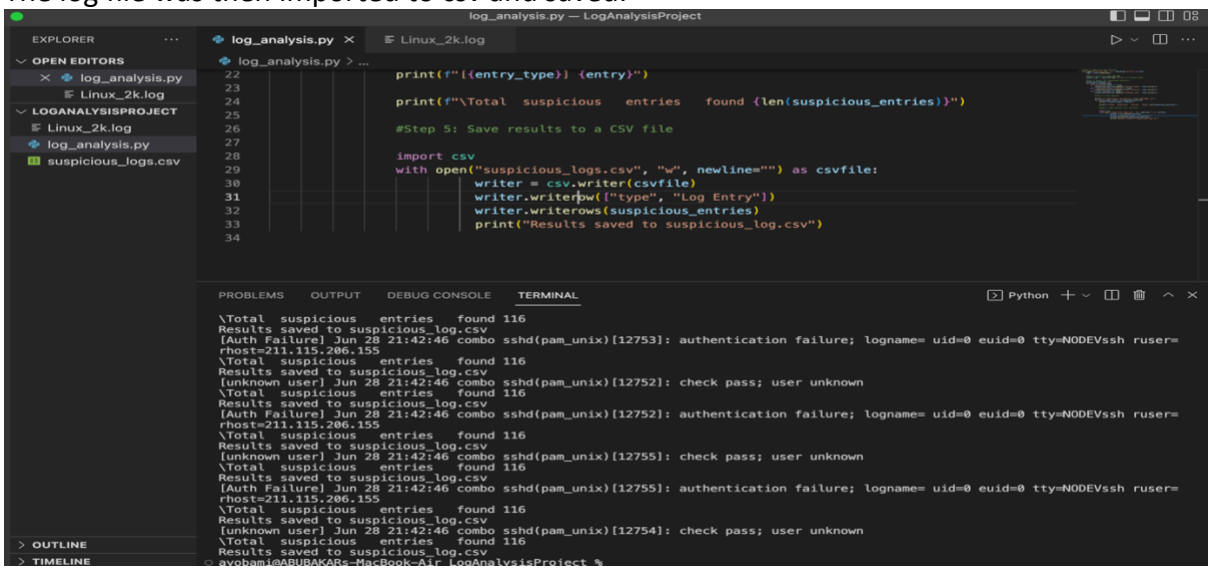


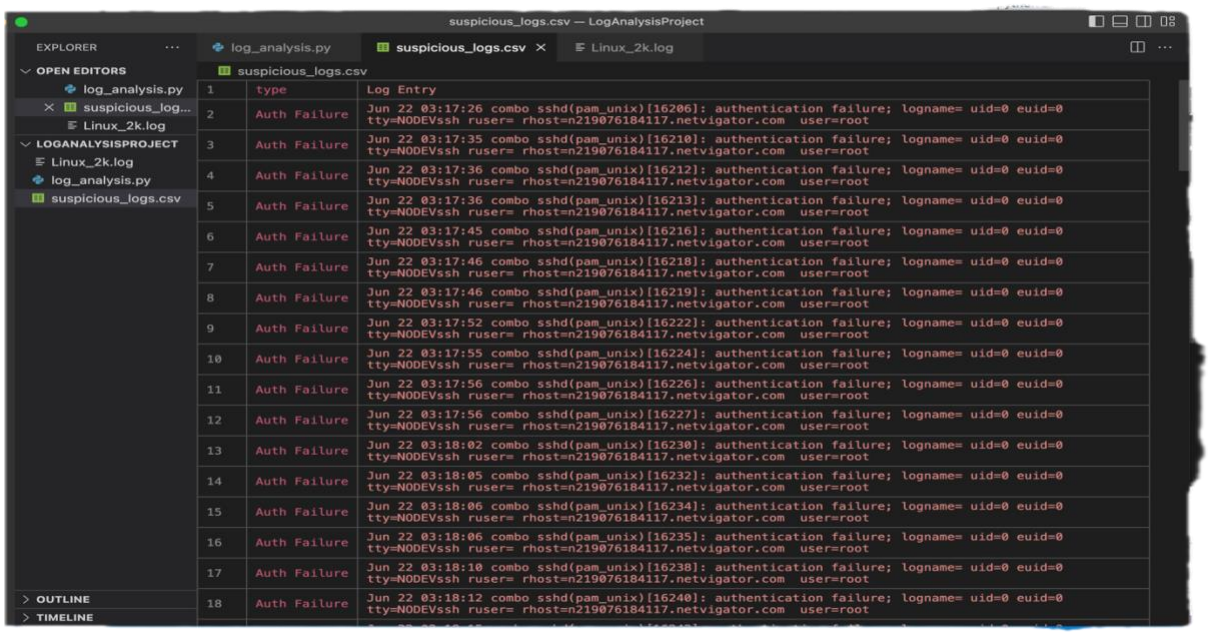
Automation:

- Developed a Python script (log_analysis.py) to streamline detection of suspicious entries.
- Script workflow: Open and read the log file.
    1. Focus on a subset of lines for targeted analysis.
    2. Search for suspicious patterns (`failed password`, `authentication failure`, `user unknown`).
    3. Aggregate and print findings, including total counts of suspicious entries.
- Result: Efficient extraction of high-frequency failed login attempts, reducing manual effort and providing structured data.



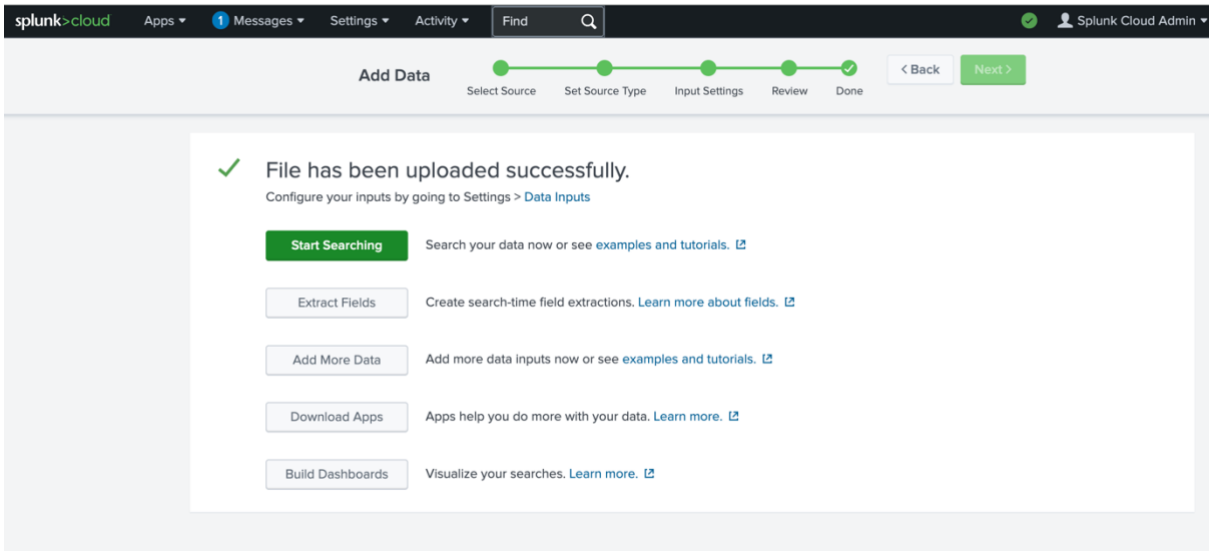
The log file was then imported to csv and saved.





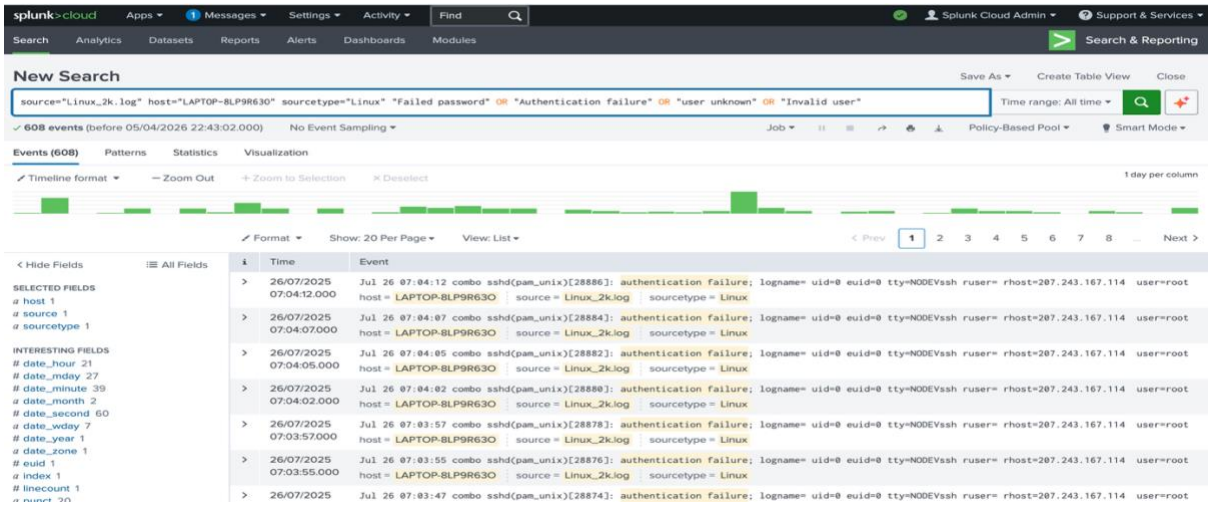
Further Investigation Analysis into the log using SIEM(Splunk)

The raw log was ingested into Splunk Cloud to perform scalable search, correlation, and visualization.



This SPL query

“ source="Linux_2k.log" host="LAPTOP-8LP9R630" sourcetype="Linux" "Failed password" OR "Authentication failure" OR "user unknown" OR "Invalid user" “

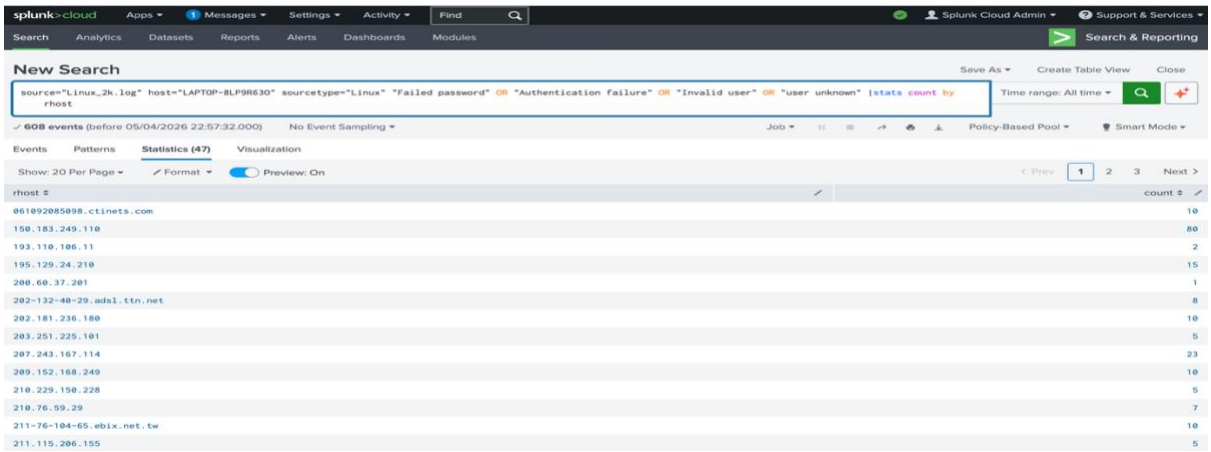


Findings:

- There are 608 events identified matching suspicious authentication patterns.

This SPL Query

“source="Linux_2k.log" host="LAPTOP-8LP9R630" sourcetype="Linux" "Failed password" OR "Authentication failure" OR "user unknown" OR "Invalid user" | stats count by rhost”

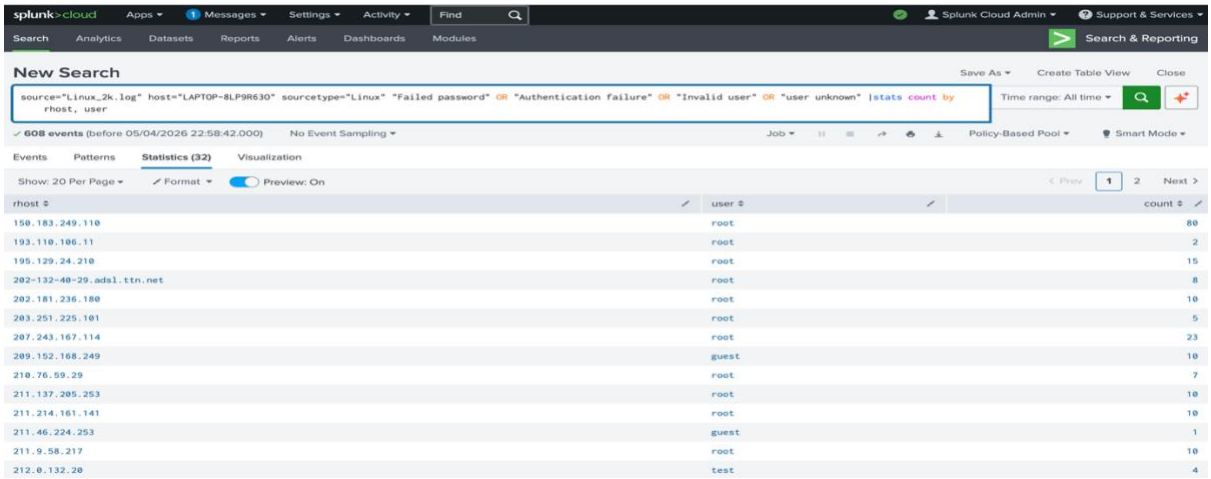


Key Findings

- 608 suspicious events detected across multiple remote hosts.
- Certain IP addresses showed high activity, e.g., one host had 80 failed login attempts, confirming automated brute-force behaviour.

This SPL Query

“source="Linux_2k.log" host="LAPTOP-8LP9R630" sourcetype="Linux" "Failed password" OR "Authentication failure" OR "user unknown" OR "Invalid user" | stats count by rhost, user”



Key Insights from Query Results

The Total suspicious events: 608 entries matched the criteria.

High-frequency attack sources:

- 150.183.249.110 attempted 80 logins targeting root.
- 207.243.167.114 attempted 23 logins targeting root.

Multiple target accounts:

- Most attacks targeted root (privileged account).
- Some attempts targeted lower-privileged or test accounts (guest, test).

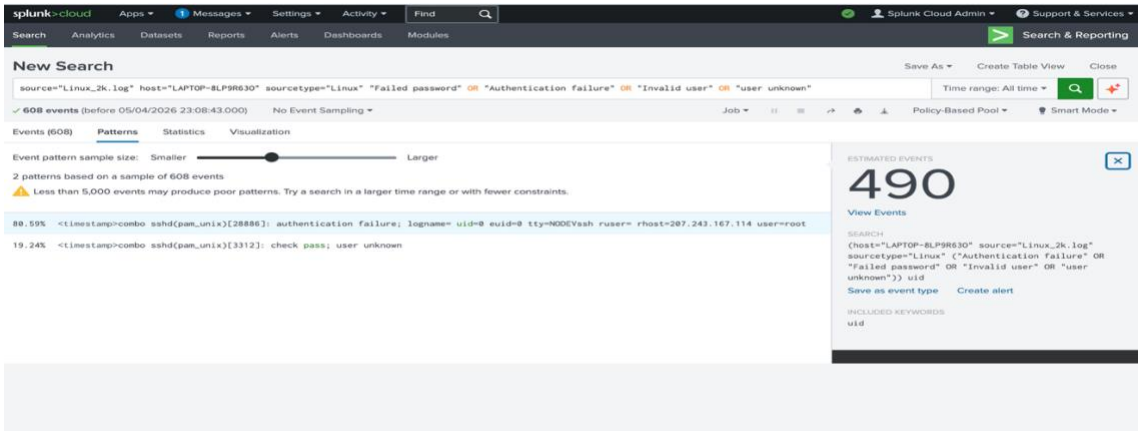
Behavioural pattern:

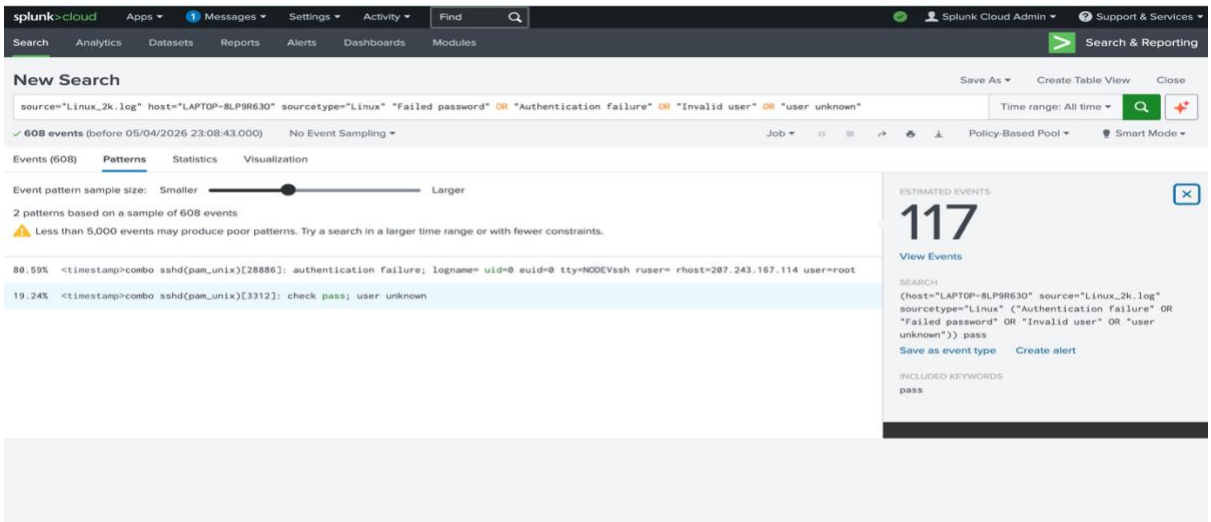
- High count of events from the same IP in a short timeframe indicates automated brute-force attacks.
- IPs with fewer attempts may indicate scanning or opportunistic probing.

Event pattern analysis indicated:

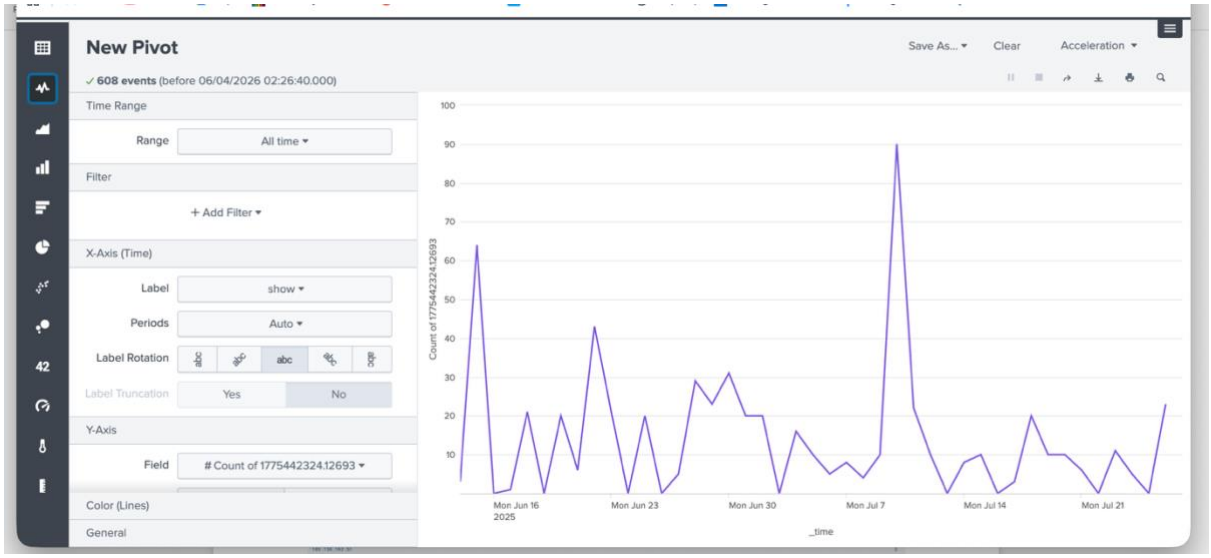
- ~80% of events were authentication failure for root user.
- ~19% were user unknown, suggesting username enumeration attempts.

Patterns confirmed non-human behaviour, consistent with automated brute-force attacks.





The Pivot visualization provides a time-series view of authentication-related events. By plotting event counts across the full-time range, the chart highlights several distinct spikes in activity, each representing concentrated bursts of failed login attempts.



Key Observations

- Multiple peaks appear throughout the timeline, with the most significant spike occurring around early July 2025.
- These peaks correlate with periods of high-frequency authentication failures, consistent with automated brute-force attempts.
- The fluctuations between low and high event counts indicate intermittent attack waves, rather than continuous manual attempts.
- The pattern aligns with behaviour typically seen in botnets or scripted SSH scanners, which operate in bursts across distributed IPs.
- The overall trend confirms that the system experienced repeated, time-clustered attack activity, supporting earlier findings from the SPL queries.

Recommendations:

Strengthen Authentication Security

- Enforce strong, complex passwords and regular password rotation for all privileged accounts (e.g., root).
- Implement multi-factor authentication (MFA) for SSH access to mitigate risks from credential-based attacks.
- Disable direct root login and enforce sudo-based privilege escalation for administrative tasks.

Limit and Monitor Remote Access

- Restrict SSH access to known and trusted IP addresses via firewall rules or VPN segmentation.
- Deploy fail2ban or equivalent intrusion prevention tools to automatically block IPs after a threshold of failed login attempts.
- Monitor authentication attempts and alert on high-frequency failed logins or unusual access patterns.

Enhance Log Management and SIEM Usage

- Develop automated detection rules and dashboards in Splunk for:
    i. Repeated authentication failures
    ii. Multiple failed attempts targeting privileged accounts
    iii. Username enumeration attempts
- Schedule regular automated log parsing with Python scripts to pre-filter suspicious events for analyst review.

Behavioural and Threat Intelligence Integration

- Correlate failed login attempts with external threat intelligence to identify malicious IP addresses or botnet activity.
- Leverage MITRE ATT&CK T1110 mapping to guide defense-in-depth strategies and detection coverage.

Incident Response and Containment

- Develop a response playbook for brute-force detection events:
    i. Immediate IP blocking for high-frequency attackers
    ii. Account lockout or password reset for targeted accounts
    iii. Post-incident analysis to identify lateral movement attempts
- Ensure timely escalation to SOC management for persistent attack campaigns.

Continuous Improvement and Training

- Conduct periodic security assessments and penetration tests to validate SSH hardening measures.
- Train system administrators and SOC analysts to recognize automated attack patterns and understand SIEM alerts.
- Incorporate lessons learned into continuous tuning of detection rules to reduce false positives and improve detection accuracy.